Scalability & Future Plan

Date	15 March 2024
Team ID	**SWTID-2026-6775**
Project Name	**FitBuddy - AI Fitness plan Generator using Gemini Models**
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Gemini API Rate Limits on Free Tier	High latency or request failure during concurrent user access.	High
2	In-Memory User Data Storage	User workout history and progress loss upon server restart.	High
3	Single Modality Input (Text Only)	Users cannot upload food/body progress images or voice prompts for plan customization.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles single/few concurrent requests using lightweight server.	Migrate to auto-scaling cloud microservices (AWS Lambda / Google Cloud Run) with Redis caching.
2	Data Storage	Local transient session storage.	Implement MongoDB / PostgreSQL database for persistent user profiles, plans, and fitness metrics.
3	Performance	Synchronous LLM responses cause waiting delay.	Implement asynchronous response streaming, background job queues (Celery/RabbitMQ), and prompt caching.
4	Security	Basic client-side API key handling or open endpoints.	OAuth2 / Firebase Authentication, encrypted DB storage, and backend proxy for Gemini API token management